HarnessBank: Semantic Gene-Bank Search with Gated Verification for Agent-Harness Self-Evolution

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EverMind AI · Shanda Group · arXiv 2607.13683v2

HarnessRSI 66 ★★

evidence: retained

Paper

1 What changes, what is kept

1 Start empty; keep executable harnesses
Cells = component × hypothesized pathology
No external task-answer corpus is preloaded.

2 Select the best training parent
Vanilla + bank → highest training utility
Read task outcomes, traces and metadata.

3 Invent or recombine patches
Edit prompts / knowledge / runtime / config
Frozen weights; evaluation kernel protected.

4 Screen on a training subset
Valid ledger + activation beacon fires
Positive paired gain and z ≥ 1.96

5 Full-train score → cell competition
Fill empty cell, or replace only if better.
Other pathology cells stay for recombination.

6 Train-selected winner → sealed test
Three attempts per task; not three search seeds.
Training admission ≠ held-out evidence.

Bank entries come from training failures, not a pre-curated retrieval set. The paper withholds test tasks but reports no related-task contamination audit.

2 Held-out gains across seven domains

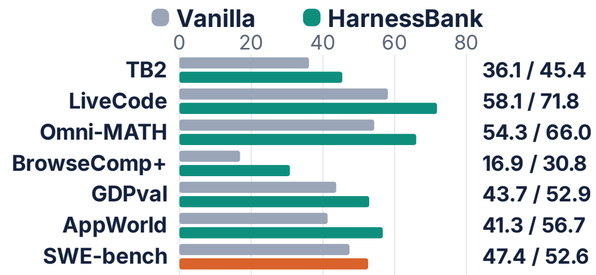


Table 1 · Test Pass@1 (%), Qwen3.6-27B, K=3 attempts/task. Six gains clear the paired bar; SWE-bench (orange) does not. No evolution-run mean/SD/CI.

4 Where the gain is absent or thin

- **SWE-bench:** 47.4 → 52.6; +5.1 reported, z=0.78, 26 test tasks. DGM: 56.4‡. Neither is credited vs vanilla.
- **Omni-MATH patch transfer:** +11.0 on Qwen397B, −1.5 on Gemini. Gemini's matched patch: +15.3 (Table 2).
- **Mechanisms concentrate:** runaway recovery dominates four of six domains; verify-finalize recurs. Main results use one task model.

‡ Figure 4 label, not computed. Transfer tests: two domains, no cross-benchmark bank transfer.

3 Controls, and what they establish

Gate ablation · TB2, Table 3

Variant	Test	False elites	Stop
Full	45.4	0	10.0
No 2σ	±0.0 Δ	+2	>20 (cap)
No confirm + 2σ	−1.6 Δ	+3	>20 (cap)

Δ relative to full. >20 (cap), as reported: stop rule unmet. No run variance or isolated bank ablation.

Protocol unclear: an inert elite passes despite the stated activation gate; the 2σ change is not cleanly isolated.

Comparable effort, not identical compute

- GEPA / DGM: 780–2,310 rollouts, within 2.1×. Same task model, splits, evaluation ruler—not internal screening.
- Five tests: credited vs vanilla on four (HarnessBank), one (DGM), none (GEPA). Not pairwise superiority tests.
- No equal-size/curation bank baseline or full optimizer token/$/time accounting reported.

Selection and protocol limits

- Training selects elites/winner; test withheld. Default-selection split and baseline tuning budget unspecified.
- Proposer parity unresolved: §4.1 names Opus 4.8; §4.3 names Qwen for baselines. No clarification.

Takeaway

Held-out lift is reported. Gate ablations suggest cleaner archives and earlier stopping, but the unexplained inert elite limits causal attribution. The bank's own contribution is not isolated.

Our read · HarnessRSI

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Reports: held-out lift, task-paired statistics, gate ablation, model transfer.

Not reported: search-seed SD/CI, isolated bank control, contamination audit, full search cost.